A Generalized Framework for the Assessment of Various Configurations of Cable-Driven Mobile Lower Limb Rehabilitation Exoskeletons

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ABSTRACT

The overall increase in the number of stroke patients and the high cost and limited accessibility to rehabilitation services have motivated the design and development of mobile exoskeletons. However, the majority of existing lower limb exoskeletons continue to be heavy, inducing unnecessary inertia and inertial vibration on the limb. Cable-driven exoskeletons can overcome these challenges without the need for exact joint alignment. The routing and configuration of cables can be accomplished in various ways while designing the exoskeleton. In this work, we propose a generalized framework for assessing the viability of cable-driven exoskeletons designs based on various routing and configurations of cables. In the current proposed framework, the lower limb has been modeled as a two-link model, and the desired trajectory is tracked via impedance control. The passive elastic joint moment is considered as user voluntary input from the impaired leg. Four conceptual models with 2-, 3- and 4- cable configurations have been generated as case studies and the resulting model trajectories were compared. Our preliminary results revealed that a 4-cable configuration is a promising design option for lower limb rehabilitation based on tracking performances, cable tensions, and component forces.

CCS CONCEPTS

• **Mathematics of computing** → Mathematical analysis; Mathematical optimization; • **Human-centered computing** → Human computer interaction (HCI); Interaction design; Systems and tools for interaction design; • **General and reference** → Cross-computing tools and techniques; Design.

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KEYWORDS

Exoskeleton, lower limb rehabilitation, cable-driven exoskeleton, link-based model, cable routing, torque mapping

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1 INTRODUCTION

Hemiparetic gait is the most common walking impairment from cerebral stroke [1] and is the result of deviations and compensatory motion-induced movement [2]. Thus, there is a need for a rehabilitation device that can support/guide the impaired limb to mimic/follow a healthy human gait pattern during walking.

Hemiparetic gait is the most common walking impairment observed in stroke patients and is characterized by asymmetry associated with an extensor synergy pattern of hip extension and adduction, knee extension, and ankle plantarflexion and inversion [1] [2]. Due to its multiple potential advantages over manual therapy, including automated, accurate, repetitive intervention, and the potential for home rehabilitation solutions, robotic rehabilitation has recently been at the forefront of stroke rehabilitation research [3–5]. Robots not only can provide task-based high intensity, repetitive functional movement training, but can also offer continuous and objective quantitative movement assessment, as well as being economical for large scale use [6].

Various types of rehabilitation devices have been proposed and tested over the last decade, including ANdROS [4], LOKOMAT [5], AIRGAIT [6], AGoRA [7] to name a few. Most of these devices operate by actuating joints directly. This may induce additional inertial vibration due to the presence of an actuator near a joint, hence increasing the load acting on the paretic limb of the patient, which is already weak and has lesser load-bearing capacity. Moreover, actuation at the joint level directly requires strong structural support

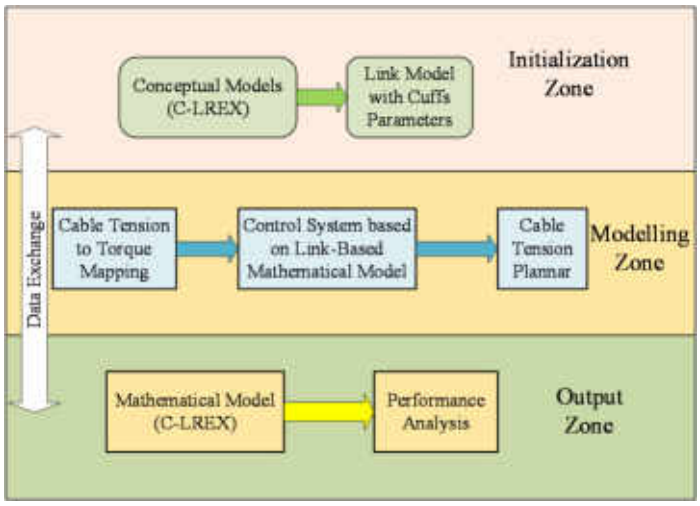

Figure 1: The generalized layout of the framework

on acting and reacting parts, which makes these systems heavy and bulky. A further common shortcoming includes pin joint assumed actuation of the knee joint, which may induce unnecessary moments/reactions [8] on the joint. Cable-driven rehabilitation devices (CDRD) can overcome the limitations of exoskeleton rehabilitation robots mentioned above. The cables, which are used as transmission elements, facilitate the remote location of the actuators, and hence minimize the inertia and inertial vibration on the impaired limb, as well as the need for strong structural support. Actuating via cables also mitigates the burden of exact joint alignment of actuating mechanisms as it acts as an end effector.

C-ALEX [9] is a cable-driven rehabilitation device that employs multiple cables to generate the desired torque on the joints. It has 3 cuffs placed on the user's lower limb (including one on the waist) to route the cables and cable ends. The robot employs a force-field controller to track the user's ankle during training. Experiments on healthy subjects showed that C-ALEX assisted users to track the desired motion of the limb with the selected ankle trajectory. Kirby et al. [10] designed and tested two cable-based lightweight (0.76 kg) knee rehabilitation exoskeletons using a powerful tethered testbed. The control systems and actuators were located on the testbed itself. The power was transmitted to the exoskeleton via a cable which allows rapid modification in the design of the exoskeleton. The device test showed its potential to be used as a knee rehabilitator. However, rehabilitating the knee joint alone is not likely to be sufficient to regain a healthy gait trajectory in stroke patients. Bryson et al. [11] analyzed 3 DOFs robot leg actuated device using 4 cables to identify the optimal parameters that are most commonly associated with performance, such as routing and cable configuration. The configurations that met the design requirements based on the analysis are then used in designing the cable-driven robot.

Aliakbar et al. [12] designed a 4 cable-driven lower limb rehabilitation exoskeleton that tacked joints and ankle position employing a proportional-derivative (PD) controller and force field controller respectively. The cables were routed from the fixed frame (attached with treadmill) to the lower limb, one cable was assigned for the hip and knee joint while the remaining cables were assigned for ankle tracking. The design constrained the overall device mobility along with the routing and configurability of cables.

Although it is well accepted that CDRD has great potential in robotic rehabilitation, the optimal cable routing(s) and configuration towards enhancing design options and users' safety and comfort remain elusive. To bridge this gap, this work purposes a generalized methodology/framework to assess the feasibility of various designs of cable-driven exoskeletons based on different routings and configurations. A general brief layout of the framework is depicted in Figure 1. Here, C-LREX stands for the Cable driven Lower limb Rehabilitation Exoskeleton, which is the proposed conceptual model for the case study.

2 METHODOLOGY

Conceptual models with different cable routings and configurations (Figure 2) are developed and converted into link-based models (Figure 3) representing all the necessary parameters (such as the limb, cable anchors, and cuffs dimensions) are identified. Based on the identified parameters, the inputs required for the controller (such as the control performance matrix) are determined. Desired joint moments are estimated employing a PD controller and transformed into equivalent cable tension via cable tension planar. The cable tension(s) are then fed into the mathematical (link model) model of the C-LREX and the predictions in terms of joint angles, joint angle velocities, joint equivalent compressive force, and joint equivalent shear force were compared to assess the feasibility of the particular configuration.

2.1 Conceptual Models

In this work, we propose four possible cable configurations (shown in Figure 2) named based on the number of cables used and their routings as 2-cables, 3-cables_short (refer to the configuration where the 3rd cable is a hip extension cable), 3-cables_long (refers to the configuration where the 3rd cable controls combined hip-knee flexion) and 4-cables configurations. Following the methodology

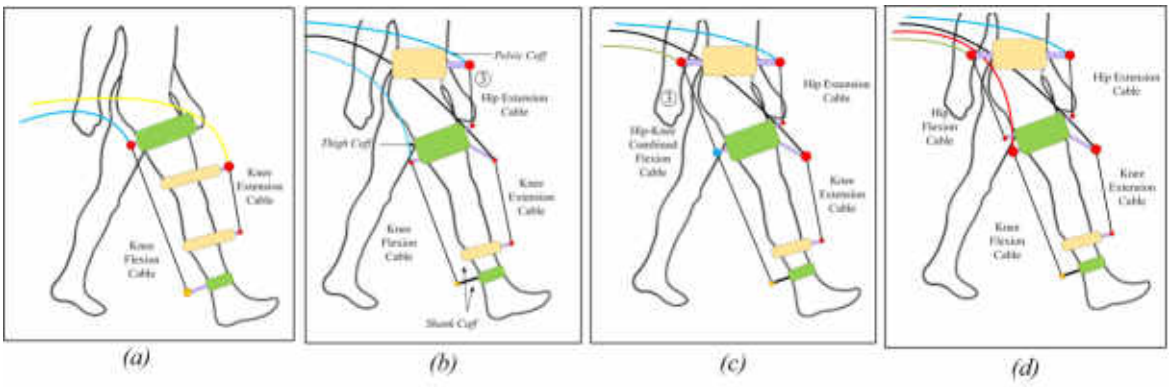

Figure 2: Proposed Conceptual Models: (a) 2- cables configuration, (b) 3-cables_short configuration, (c) 3-cables_long configuration, and (d) 4-cables actuated configurations of the model.

Table 1: Routing parameters description

Name	Type	Description
hx_fix	Distance (*m*)	Hip Joint to Pelvis cuff (Horizontal axis)
hy_fix		Hip Joint to Pelvis cuff (Vertical axis)
c_lh_A		Hip Joint to Thigh cuff (for Hip Joint) along thigh axis
a_lh_A		Thigh to Thigh cuff (for Hip Joint) anchor at a certain angle (*t_h_A*)
c_lh		Hip Joint to Thigh cuff (for Knee Joint) along thigh axis
a_lh		Thigh to Thigh cuff (for Knee Joint) anchor at a certain angle (*t_h*)
c_ln		Knee Joint to Shank cuff anchor along thigh axis
a_ln		Knee Joint to Shank cuff anchor at a certain angle (*t_n*)
t_h	Angle (*Deg*)	Thigh to Thigh cuff anchor
t_h_A		Thigh to Thigh cuff anchor
t_n		Shank to Shank cuff anchor

(Figure 1), the proposed conceptual model's feasibility for the model is analyzed by comparing their tracking performances and other parameters such as cable tension(s), shear, and compressive force on the joints due to cable tensions. For simplicity, backside cables are denoted as flexion cables, while front side cables are labeled as extension cables for both the hip and knee joints (taking the knee as reference).

2.2 Generalized Cable Routing Framework

To accommodate the proposed conceptual models for analysis in a similar framework, a generalized framework for cable routing was defined. The framework can accommodate a variety of possible cable routing configurations by changing parameters, including cuff length, distance from the joints, and the angular inclination from the attached limb. In this framework, the ankle is assumed to be fixed to the shank perpendicularly. The different parameters describing the cable routing are summarized in Table 1. The angles are taken as positive in the clockwise direction and the respective limb is assumed as reference. In Figure 3, F and E refer to Flexion and Extension (in reference to the Knee joint), or back and front side, respectively.

3 MODELING OF LIMB

Based on the link model [6, 9, 12, 13], we considered the lower limb as a two-link model connected to the pelvis at the hip and knee joints, respectively (Figure 4). The use of cables facilitates the remote location of actuators and requires lighter anchor points as compared to rigid link-based systems. The link model adopted here is identical to a double pendulum model assuming negligible weight and inertia of the cuffs. The dynamic equation of motion can therefore be formulated using Newton or Euler's- Lagrange equations, using $\theta = [\ \theta_1 \quad \theta_2\ ]^T$ as the generalized coordinate, where θ_1 and θ_2 represent the hip and knee joint angular positions measured from the vertical axis, respectively.

The generic dynamic equation of the model can be expressed as:

$$M(\theta)\ddot{\theta} + C\left(\theta, \dot{\theta}\right)\dot{\theta} + G(\theta) = \tau \tag{1}$$

Where $M(\theta)$ represents the inertial matrix (2×2) $C(\theta, \dot{\theta})$ represents the Coriolis component (2×1), $G(\theta)$ represents the Gravitational components (2×1), and τ represents the torques on the joints.

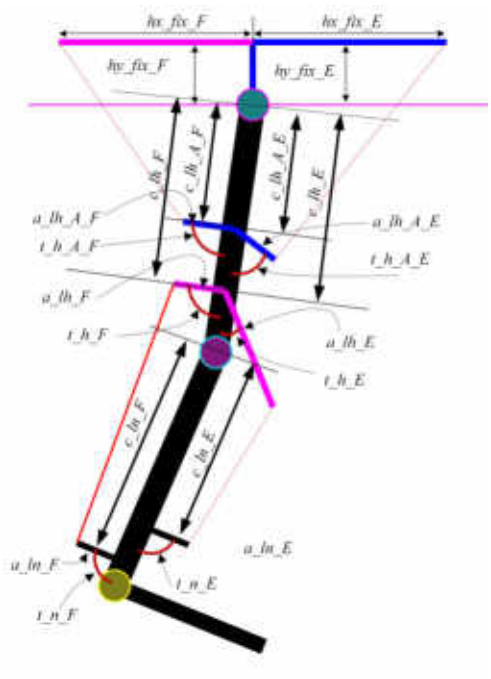

Figure 3: Generalized Cable Routing Framework

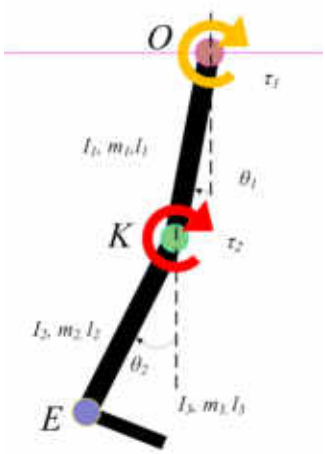

Figure 4: Two link-based lower limb model

3.1 Force to Torque Mapping

The proposed conceptual models employ cables to generate the required torque at the hip and/or knee joints. To generate the desired torque at a given joint and configuration, the relationship between the cable tension and the torque generated can be derived by either using the principle of virtual work (as derived in work [9]) or via vector algebra methods. Though each method yields a similar output, employing the vector algebra approach provides the tangential and normal forces acting at the joints while calculating the torques induced via the cables. In Figure 5. F_{tan} and F_{nor} represent the compressive and shear forces respectively induced on the joints. Subscripts *1* and *2* represent the hip and knee joints respectively.

The projection of the cable tension vector ($\overrightarrow{EA}$) along the shank vector provides the tangential component of the force, while the orthogonal component of ($\overrightarrow{EA}$) provides the normal force component. The torque on the joint (caused by the cable tension) can be estimated using the cross-product of the distance vector and the force vector. Using the same principle, the force components of the shank can be found as:

$$\begin{aligned}
\overrightarrow{F_{\tan 2}} &= \left(proj_{\widehat{EK}}\widehat{EA}\right) = \left(\frac{\left(\widehat{EA}\bullet\widehat{EK}\right)}{\left\|\widehat{EK}\right\|^2}\widehat{EK}\right) \\
\overrightarrow{F_{nor2}} &= \left(\widehat{EA} - proj_{\widehat{EK}}\widehat{EA}\right) = \left(\widehat{EA} - \frac{\left(\widehat{EA}\bullet\widehat{EK}\right)}{\left\|\widehat{EK}\right\|^2}\widehat{EK}\right) \\
\overrightarrow{\tau_2} &= \left(\overrightarrow{KE} \times \widehat{EA}\right)
\end{aligned} \tag{2}$$

Where, $\widehat{EA}$ *and* $\widehat{EK}$are the unit vector along the $\overrightarrow{EA}$ *and* $\overrightarrow{EK}$ direction respectively.

Similarly, the equivalent force components and torque on the hip joint can be estimated as:

$$\begin{aligned}
\overrightarrow{F_{\tan 1}} &= \left(proj_{\widehat{KO}}\overrightarrow{F_{\tan 2}}\right) + \left(proj_{\widehat{KO}}\overrightarrow{F_{nor2}}\right) + \left(proj_{\widehat{KO}}\widehat{AE}\right) \\
\overrightarrow{F_{nor1}} &= \overrightarrow{F_{\tan 2}} - \left(proj_{\widehat{KO}}\overrightarrow{F_{\tan 2}}\right) + \overrightarrow{F_{nor2}} - \left(proj_{\widehat{KO}}\overrightarrow{F_{nor2}}\right) \\
&\quad + \widehat{AE} - \left(proj_{\widehat{KO}}\widehat{AE}\right) \\
\overrightarrow{\tau_1} &= \left(\overrightarrow{OK} \times \overrightarrow{F_{\tan 2}}\right) + \left(\overrightarrow{OK} \times \overrightarrow{F_{nor2}}\right) + \left(\overrightarrow{OA} \times \widehat{AE}\right)
\end{aligned} \tag{3}$$

In the above equations, the actual value of forces and torque are obtained by multiplying with the magnitude of the cable tension.

In equations 2) and (3), the unknown for a given cable routing configuration is the force magnitude. Thus, the relation between torque and force can be written as:

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} F = B^T F \tag{4}$$

Where, $B^T = |\ B_1 \quad B_2\ |^T$ is the control performance matrix that maps the cable tension F to the hip joint and knee joint torque.

3.2 Control System

Based on the 2-link model, a three-layer control algorithm was implemented to track the desired trajectory during motion, similar to the generalized control framework for the orthosis (impedance control) in [4], [14], [15]. Figure 6 depicts the generalized control strategy with the all-control layers.

The system is divided into upper and lower control layers, as well as the data layer. The upper layer is responsible for generating the desired torque required (using a PD controller) for tracking the next step based on the current step data. The data layer estimates the inertial and other state parameters (like *M, C, G)* of the system at a given instant of the motion. The data from the data layer is fed into both upper and lower control layers. The lower control layer, based on the data from both the upper layer and data layer, allocates cable tensions to the individual cables.

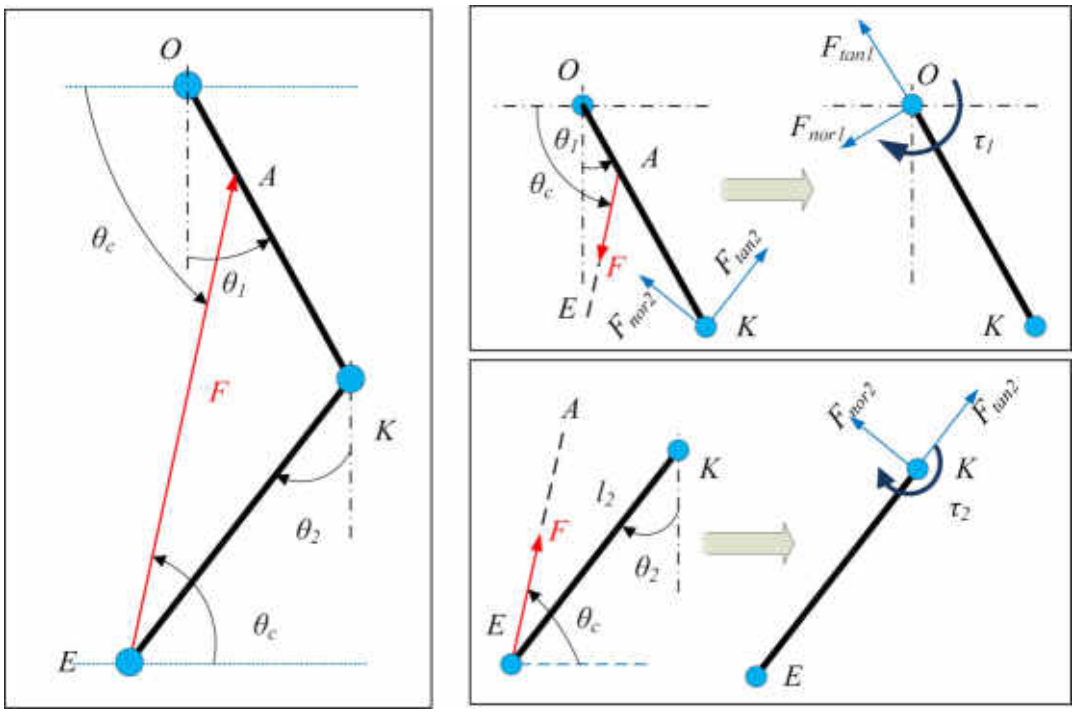

Figure 5: Two link model with applied cable tension (Left), Force to Torque mapping for shank (Right Bottom), and Force to Torque mapping for thigh (Right Top)

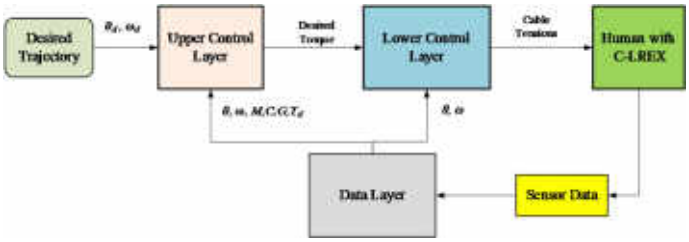

Figure 6: Three-layer control system

3.3 Cable Tension Planar

In the previous section, the force to torque mapping facilitated in estimating the toque being exerted on the joint due to the cable tension(s). However, the adopted mathematical model and control architecture estimate the required torques to be applied on the joint during trajectory tracking. Thus, the lower control layer requires a reverse mapping in order to map desired torque to the desired cable tension(s).

Based on equation 4), the control performance matrix (B) could be invertible or non-invertible depending on the number of cables and routing configurations. Thus, an exact solution to estimate the cable tension F is not possible for most configurations.

Equation 4) can be written as a generic optimization problem as:

$$\min_{F} \left(J\left(F\right)\right) \quad s.t. \begin{cases} \tau = BF \\ F_{\min} \leq F \leq F_{\max} \end{cases} \tag{5}$$

The cost function $J(F)$ can be selected as a hybrid optimization cost function that includes a weighted combination of the cost function for error minimization and the cost function for control minimization. To guarantee a feasible unique solution in each case, the equation 5) is rearranged with the hybrid cost function to convert the optimization problem into a quadratic programming problem similar to the control allocation problem in work [16].

$$\min_{F} \left(\tfrac{1}{2}F^T HF + f^T F\right) \quad s.t. \ \{F_{\min} \leq F \leq F_{\max} \tag{6}$$

4 SIMULATION RESULTS AND DISCUSSION

In the current study, it is assumed that the stroke-affected leg has no active muscle contribution in walking (only muscle contribution is due to the passive elastic joint moment [17]). The passive elastic joint moment consists of moments generated due to the stretch of soft tissues and ligaments around the joint. The anthropometric data for the lower limb is adopted from Winter's [18] model. The reference gait trajectories for the hip and knee joints in the sagittal plane are extracted from Bovi et al. [19]. A complete gait cycle was tracked assuming the leg to be suspended in the air (air gaiting) since the device is intended to support the swing phase of the gait. For the cable tension-planning module, the minimum cable tension was set to ***7N*** to ensure that all cables are always in tension and the maximum was set at ***150N*** and ***350N*** for knee and hip cables respectively.

Based on the assumption and simplifications stated in the above section, a *MATLAB* model was developed and simulated to track

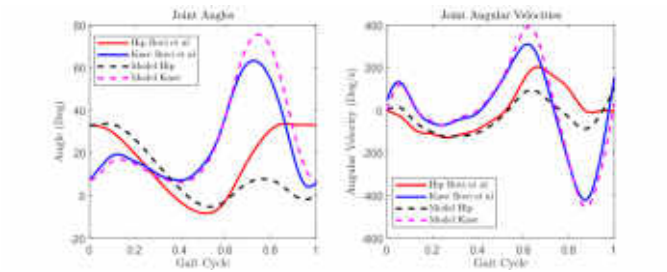

Figure 7: Tracking performance with 2-cables

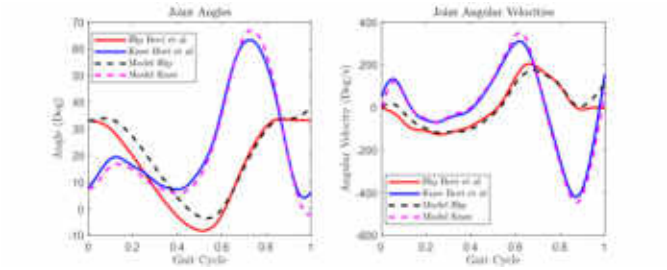

Figure 8: Tracking performance with 3-cables_short configurations

the healthy gait trajectories for the hip and knee joints, and the corresponding required cable tensions were analyzed.

4.1 Tracking with 2 cables

Figure 7 depicts the tracking performance with 2 cables configuration. The figure suggests that stroke patients with negligible muscle capacity in the impaired limb fail to follow the desired gait trajectory with 2 cables. The 2-cable configuration can only generate relative torque on the knee joint but not on the hip joint which leads to poor tracking performance.

4.2 Tracking with 3 cables_short configuration

The tracking performances with the 3-cables_short configuration is shown in Figure 8. This configuration partially tracks the desired joint trajectory. Though the additional cable provided on the hip joint generates torque for the hip, the torque is only generated in one direction and hence the tracking deteriorates due to failure to generate the opposite direction torque by the hip cable.

4.3 Tracking with 3 cables_long configuration

Tracking performance with 3-cables_long configuration is acceptable as compared with the previous models (Figure 9). Although only one cable is employed to generate equivalent torques on both the hip and knee joints from the backside (named as flexion cable), sufficient moment is generated on both hip and knee joints. Thus, apparently, this configuration acts as 4 cables (2 individual cables for each joint), with equal tension in the backside hip and knee cable.

4.4 Tracking with 4 cables configuration

The tracking performance with 4-cables is similar to the tracking performance of the 3 cables_long configurations (Figure 10). A beneficial aspect of the 4 cables configuration is that the hip and knee cable tensions are controlled independently. Failure to exactly track the knee angle during the last phase of the gait cycle is due to the high angular velocity of the shank in the swing phase, which requires a higher torque to overcome. Since we limited the maximum allowable tension in the cables, the torque requirement at the last stage of the cycle was not met.

4.5 3 cables_short VS. 4 cables configuration

The slight difference in tracking performance observed between these two configurations leads to some difficulty in defining it as a **'better'** configuration, and thus other parameters, including the range of cable tensions, force components induced at joints, are analyzed. Figure 11 depicts the layouts of the cable tension in both cases of tracking. The cable tension in the knee flexion cable in the 4-cables configuration case is much smaller than the 3 cable_long configuration case at the start and end of the gait cycle and it remains lower throughout the gait cycle. Similarly, the knee extension cable tension pattern has a lower value at the start and end of the gait cycle for the 4-cables configuration and remains similar throughout the gait cycle. However, there is an insignificant variation in the hip extension cable tension patterns in both cases. The application of higher knee flexion cable tension (as a result of using a combined hip-knee flexion cable) not only resulted in higher knee flexion cable tension but also higher knee extension cable tension.

Figure 12 represents the compressive and shear forces respectively, induced on the joints in 3-cables_long and 4-cables configurations. It can be observed that 3 cable_long configuration induces a much higher value of compressive & shear forces on the knee joint and the higher shear force on the hip joint at the start and end of the Gait Cycle.

The reason is that the cable tension in the backside of the cable (combined hip-knee flexion) of the 3-cables_long configuration is usually higher in order to meet the demand at the hip joint. Based on the comparison between the cable tension and component forces on the joints, the 4 cables configuration is a promising candidate for the exoskeleton design.

5 CONCLUSIONS AND FUTURE WORK

Cable driven exoskeletons promise to reshape the field of robotic rehabilitation as they are lightweight, do not require exact joint alignment, and exert negligible inertia and inertial vibration on the impaired limb. The optimal number of cables and cable configurations to attain these benefits, however, remain elusive. This study proposed a generalized framework for the feasibility assessment of various configurations of the cable driven exoskeletons. A simple two-link model was used with the ankle at a fixed angle to the shank. Various configurations were simplified and transformed into MATLAB simulations via link-based modeling. The impaired limb was assumed to generate only passive elastic joint moments during motion and a three-layer controller was used for tracking the

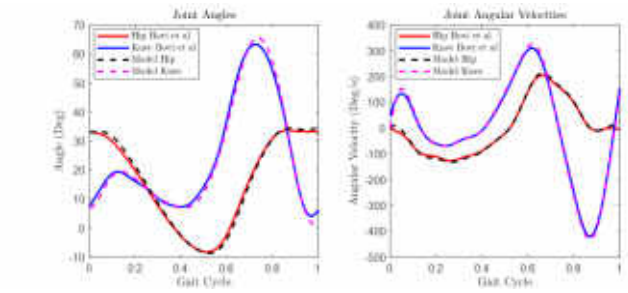

Figure 9: Tracking performance with 3-cables_long configuration

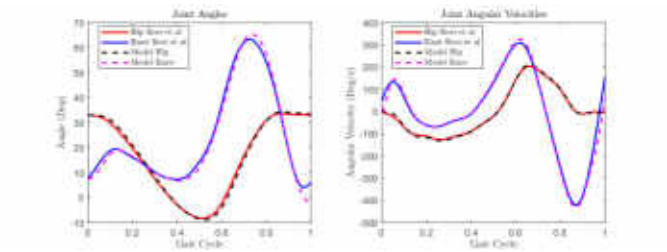

Figure 10: Tracking performance with 4-cables configurations

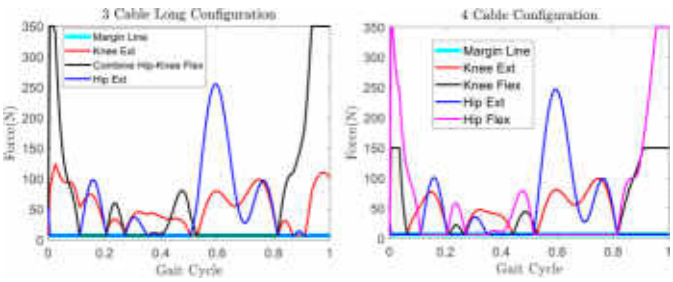

Figure 11: Cable tensions for 3-cables_long and 4-cables configurations

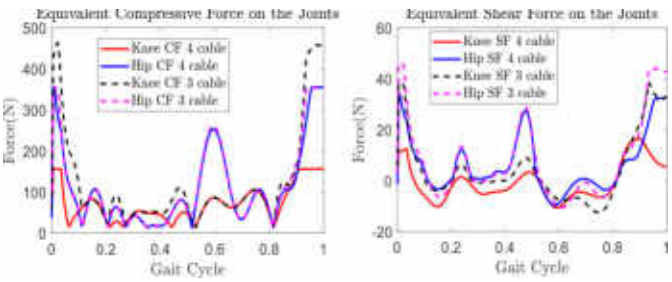

Figure 12: Comparison of compressive and shear forces induced on joints

desired healthy limb trajectory. A quadratic programming-based optimization problem was formulated to distribute the cable tensions to each cable.

The framework can accommodate broader numbers of designs for assessment by varying the parameters of cable routing and configurations (Figure 3). As a case study, 4 different typical configurations of cable driven exoskeletons were assessed as potential stroke patient rehabilitation tools. The best tracking performance was found with 4 cables configuration (reasonable against 3 cables_long configurations and better than other configurations) with lower compressive/shear forces introduction on the joints. This model is currently limited to the sagittal plane and provides a proof of concept of the value of using simulation models to study cable configuration. Future work includes the extension of the current work to accommodate knee and hip abduction/adduction and cable routings in the frontal plane.

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